

# Apollo Diagnostics - Laboratory Investigation Report

123, Nungambakkam High Road, Chennai - 600034 | NABL Accredited | Tel: 044-2827-5000

|                  |                  |              |                      |
|------------------|------------------|--------------|----------------------|
| Patient Name:    | Ramesh Kumar     | Report ID:   | APL-DIAB-52-RAM-2026 |
| Age / Gender:    | 52 Yrs / Male    | Category:    | Type 2 Diabetes      |
| Weight / Height: | 88 kg / 170 cm   | BMI:         | 30.4 kg/m2           |
| Ref. Doctor:     | Dr. Sunita Reddy | Report Date: | 18-Jan-2026          |

## 1. COMPLETE BLOOD COUNT [Circulatory System]

### Red Cell Indices

| Test                     | Result | Unit | Reference Range | Flag   |
|--------------------------|--------|------|-----------------|--------|
| Haemoglobin (Hb)         | 14.2   | g/dL | 13.0 - 17.5     | Normal |
| Packed Cell Volume (PCV) | 43.0   | %    | 40 - 52         | Normal |
| MCV                      | 90.0   | fL   | 83 - 101        | Normal |
| MCH                      | 29.0   | pg   | 27 - 32         | Normal |
| RDW-CV                   | 13.0   | %    | 11.6 - 14.0     | Normal |

**Haematology Comment:** Haemoglobin 14.2 g/dL is within normal range. Red cell indices are unremarkable.

### White Cells & Platelets

| Test           | Result | Unit      | Reference Range | Flag   |
|----------------|--------|-----------|-----------------|--------|
| Total WBC      | 7000   | cells/cmm | 4500 - 11000    | Normal |
| Neutrophils    | 62     | %         | 40 - 75         | Normal |
| Lymphocytes    | 28     | %         | 20 - 45         | Normal |
| Platelet Count | 2.8    | lakh/cmm  | 1.5 - 4.0       | Normal |

**Comment:** WBC and platelet counts within reference limits. No significant leucocytosis, leucopenia or thrombocytopenia.

## 2. IRON STUDIES [Haematopoietic System]

| Test                   | Result | Unit  | Reference Range | Flag   |
|------------------------|--------|-------|-----------------|--------|
| Serum Iron             | 110    | ug/dL | 60 - 170        | Normal |
| TIBC                   | 310    | ug/dL | 250 - 370       | Normal |
| Transferrin Saturation | 35.0   | %     | 20 - 50         | Normal |
| Serum Ferritin         | 80.0   | ng/mL | 30 - 300        | Normal |

**Iron Studies Comment:** Iron stores and transport markers are within normal limits. No iron deficiency or iron overload.

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## 3. HEPATIC FUNCTION TEST [Hepatobiliary System]

### Liver Enzymes

| Test                       | Result | Unit  | Reference Range | Flag   |
|----------------------------|--------|-------|-----------------|--------|
| SGPT / ALT                 | 22     | U/L   | 7 - 56          | Normal |
| SGOT / AST                 | 20     | U/L   | 10 - 40         | Normal |
| Alkaline Phosphatase (ALP) | 70     | U/L   | 44 - 147        | Normal |
| Total Bilirubin            | 0.6    | mg/dL | 0.2 - 1.2       | Normal |
| Albumin                    | 3.6    | g/dL  | 3.5 - 5.0       | Normal |

**Hepatologist Note:** Liver enzymes within normal limits. No hepatocellular damage or cholestatic process detected.

## 4. RENAL FUNCTION TEST [Urinary System]

| Test                | Result | Unit          | Reference Range | Flag   |
|---------------------|--------|---------------|-----------------|--------|
| Serum Creatinine    | 1.0    | mg/dL         | 0.7 - 1.3       | Normal |
| Blood Urea Nitrogen | 14     | mg/dL         | 7 - 20          | Normal |
| eGFR (CKD-EPI)      | 81     | mL/min/1.73m2 | 60 - 120        | Normal |
| Serum Uric Acid     | 4.8    | mg/dL         | 2.4 - 7.0       | Normal |
| Sodium (Na+)        | 139    | mEq/L         | 135 - 145       | Normal |
| Potassium (K+)      | 4.0    | mEq/L         | 3.5 - 5.1       | Normal |

**Nephrologist Note:** eGFR 81 mL/min/1.73m2 - glomerular filtration well preserved. Renal function normal.

## 5. LIPID PROFILE [Cardiovascular Risk]

| Test              | Result | Unit  | Reference Range | Flag |
|-------------------|--------|-------|-----------------|------|
| Total Cholesterol | 216    | mg/dL | 0 - 200         | HIGH |
| HDL Cholesterol   | 37     | mg/dL | 40 - 999        | LOW  |
| LDL Cholesterol   | 162    | mg/dL | 0 - 100         | HIGH |
| Triglycerides     | 268    | mg/dL | 0 - 150         | HIGH |

**Cardiologist Note:** LDL 162 mg/dL (target <100 mg/dL) and Total Cholesterol 216 mg/dL are elevated. Triglycerides 268 mg/dL. Increased cardiovascular risk. RECOMMENDATION: Mediterranean diet, aerobic exercise, statin therapy consideration if >10-year CV risk >7.5%.

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## 6. THYROID FUNCTION TEST [Endocrine System]

| Test    | Result | Unit   | Reference Range | Flag   |
|---------|--------|--------|-----------------|--------|
| TSH     | 2.1    | uIU/mL | 0.4 - 4.0       | Normal |
| Free T3 | 3.2    | pg/mL  | 2.3 - 4.2       | Normal |
| Free T4 | 1.1    | ng/dL  | 0.8 - 1.8       | Normal |

**Endocrinologist Comment:** Thyroid function normal. TSH 2.1 uIU/mL. Euthyroid state confirmed.

## 7. DIABETES SCREENING [Endocrine - Pancreas]

| Test                        | Result | Unit  | Reference Range | Flag |
|-----------------------------|--------|-------|-----------------|------|
| Fasting Blood Glucose       | 192    | mg/dL | 70 - 100        | HIGH |
| Post-Prandial Glucose (2hr) | 243    | mg/dL | 0 - 140         | HIGH |
| HbA1c                       | 7.3    | %     | 0 - 5.7         | HIGH |
| HOMA-IR                     | 5.6    |       | 0 - 2.5         | HIGH |

**Endocrinology - CRITICAL FINDING:** HbA1c 7.3% and FBG 192 mg/dL confirm DIABETES MELLITUS TYPE 2. PPG 243 mg/dL. Immediate antidiabetic therapy (Metformin 500-1000 mg) and dietary modification required. Quarterly HbA1c monitoring and annual funduscopy/nephropathy screening advised.

## 8. VITAMINS & BONE [Nutritional / Skeletal]

| Test              | Result | Unit  | Reference Range | Flag   |
|-------------------|--------|-------|-----------------|--------|
| Vitamin D (25-OH) | 14.1   | ng/mL | 30 - 100        | LOW    |
| Vitamin B12       | 450    | pg/mL | 211 - 911       | Normal |
| Folate (Serum)    | 12.0   | ng/mL | 5.4 - 999       | Normal |
| Calcium (Total)   | 9.4    | mg/dL | 8.5 - 10.5      | Normal |
| PTH               | 32     | pg/mL | 15 - 65         | Normal |

**Nutritionist/Endocrinologist Comment:** Vitamin D: 14 ng/mL - deficient (10-20). PTH at 32 pg/mL is within range. Supplementation: 60,000 IU cholecalciferol weekly x 8 weeks, then 1000-2000 IU/day. Recheck in 3 months.

## 9. INFLAMMATORY MARKERS [Immune System]

| Test   | Result | Unit  | Reference Range | Flag   |
|--------|--------|-------|-----------------|--------|
| ESR    | 43     | mm/hr | 0 - 15          | HIGH   |
| CRP    | 5.0    | mg/L  | 0 - 5           | Normal |
| hs-CRP | 2.3    | mg/L  | 0 - 1.0         | HIGH   |

**Immunology Comment:** CRP 5.0 mg/L and ESR 43 mm/hr are elevated. hs-CRP 2.3 mg/L - intermediate CV risk (1-3). Inflammatory markers elevated - correlate with clinical symptoms and autoimmune screen.

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## CLINICAL SUMMARY & RECOMMENDATIONS

|             |                                                                                                       |
|-------------|-------------------------------------------------------------------------------------------------------|
| [CRITICAL]  | Diabetes Mellitus Type 2 confirmed: HbA1c=7.3%, FBG=192 mg/dL. Antidiabetic therapy required.         |
| [ELEVATED]  | LDL=162 mg/dL (HIGH). Total Cholesterol=216. Dyslipidaemia - dietary + pharmacological management.    |
| [DEFICIENT] | Vitamin D=14 ng/mL (LOW). Supplementation required.                                                   |
| [ACTION]    | Next appointment: Dr. Sunita Reddy in 4-8 weeks for follow-up and review of supplementation response. |

Report reviewed by: Dr. Anitha Krishnamurthy, MD (Pathology) | Verified by Dr. Sunita Reddy

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